

PartnerAPI

Public Hotel Booking API

Software Requirements Specification

Software Engineering Project

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1 Introduction

1.1 Purpose

This Software Requirements Specification defines the functional, non-functional, interface, security, operational and quality requirements for PartnerAPI, a public API programme for a travel booking company.

The API will allow partner travel agencies to search hotels and make hotel bookings programmatically without directly logging into the company's website.

1.2 Project Background

The company intends to provide API access to 40 partner agencies.

The first five partners are targeted for production within 10 weeks. The remaining partners will follow after the initial launch.

Two partners have requested custom features that are not currently required by the wider partner population. These requests will therefore be subjected to formal change control.

1.3 Objectives

The system shall:

1. Provide secure programmatic hotel search.
2. Provide secure hotel booking capabilities.
3. Provide partner authentication and authorisation.
4. Provide a sandbox environment.
5. Provide API documentation.
6. Enforce partner rate limits.
7. Provide usage reporting.
8. Support controlled partner onboarding.
9. Maintain an API error rate below 0.5%.
10. Keep onboarding time below 5 working days per partner.

2 Scope

2.1 In Scope

- Partner registration
- Partner authentication
- API key management

- Hotel search
- Hotel availability
- Room availability
- Booking creation
- Booking retrieval
- Booking cancellation
- Rate limiting
- Access control
- Sandbox
- API documentation
- Usage reporting
- Partner onboarding

2.2 Out of Scope

- One-off partner-specific business rules
- Custom UI applications for individual partners
- Partner-specific booking workflows
- Unapproved third-party integrations
- Features without general partner value

3 Stakeholders

Stakeholder	Responsibility / Interest
Partner Agencies	Integrate with the public API and perform hotel operations.
Sales Team	Represents partner commitments and commercial requirements.
Engineering Team	Designs, implements, secures and maintains the API.
Support Team	Handles partner integration and operational support.
Project Management	Controls scope, schedule, resources and risks.
Business Management	Approves scope and launch decisions.

4 Functional Requirements

ID	Requirement	Description	Priority
FR-01	Partner Registration	The system shall allow an authorised administrator to register a partner agency.	Must
FR-02	Authentication	The system shall authenticate API requests using valid partner credentials.	Must
FR-03	Hotel Search	The system shall allow an authenticated partner to search hotels using supported destination and stay parameters.	Must
FR-04	Availability	The system shall return available hotels and rooms matching a valid search request.	Must
FR-05	Booking Creation	The system shall allow an authenticated partner to create a hotel booking for an available room.	Must
FR-06	Booking Retrieval	The system shall allow a partner to retrieve booking information for authorised bookings.	Must
FR-07	Booking Cancellation	The system shall allow a partner to request cancellation of an eligible booking.	Must
FR-08	API Key Management	The system shall allow authorised administrators to issue, revoke and rotate partner API credentials.	Must
FR-09	Rate Limiting	The system shall enforce partner-specific API request limits.	Must
FR-10	Sandbox	The system shall provide a non-production environment for partner integration testing.	Must
FR-11	Usage Reporting	The system shall record API usage information and provide authorised users with usage reports.	Should
FR-12	Documentation	The system shall provide API documentation describing endpoints, authentication, request formats, responses and error handling.	Must

5 Non-Functional Requirements

ID	Category	Measurable Requirement	Priority
NFR-01	Latency	At least 95% of valid hotel-search requests shall receive a response within 500 ms under the agreed normal production test load.	Must
NFR-02	Error Rate	The production API shall maintain an application error rate below 0.5% over a rolling operational monitoring period.	Must
NFR-03	Availability	The production API shall achieve at least 99.5% monthly availability excluding approved maintenance windows.	Should
NFR-04	Security	All production API communication shall use TLS 1.2 or higher.	Must

ID	Category	Measurable Requirement	Priority
NFR-05	Authentication Security	Invalid or revoked partner credentials shall not be accepted for API access.	Must
NFR-06	Rate Limiting	When a partner exceeds its configured request limit, the API shall reject the excess request with HTTP 429 and shall provide an appropriate rate-limit response.	Must
NFR-07	Onboarding	The average time required to onboard an approved partner shall remain below 5 working days.	Must
NFR-08	Documentation	All production API endpoints shall have documented request parameters, authentication requirements, success responses and error responses before release.	Must

6 External Interface Requirements

6.1 API Interface

The public API shall expose versioned endpoints.

Example endpoints include:

`/api/v1/hotels/search`

`/api/v1/bookings`

`/api/v1/bookings/{bookingId}`

`/api/v1/bookings/{bookingId}/cancel`

6.2 Authentication Interface

Partners shall provide valid API credentials with API requests.

Protected endpoints shall reject requests containing invalid, expired or revoked credentials.

6.3 Error Interface

The API shall provide machine-readable error responses containing:

- HTTP status code
- Error code
- Human-readable message
- Request or correlation identifier where applicable

7 Security Requirements

The system shall:

- Authenticate every protected API request.
- Reject invalid credentials.
- Support credential revocation.
- Protect communication using TLS.
- Apply partner-specific access control.
- Apply rate limits.
- Avoid exposing internal system credentials.
- Log security-relevant events.

8 Constraints

1. The first five partners are targeted for launch within 10 weeks.
2. Five working days constitute one working week.
3. The provided three-point estimates shall be used for project estimation.
4. The public API shall not expose internal booking implementation details.
5. Custom partner requests shall follow change control.

9 MoSCoW Prioritisation

9.1 Must Have

- Authentication
- Hotel search
- Availability
- Booking
- Cancellation
- API key management
- Rate limiting
- Sandbox
- Documentation
- Security controls

9.2 Should Have

- Usage reporting
- Advanced operational reporting

9.3 Could Have

- Additional analytics
- Additional partner self-service features

9.4 Won't Have in Initial Release

- Partner-specific custom business logic
- Individual partner UI applications
- Unapproved one-off integrations

10 Assumptions

- The internal booking system provides the necessary hotel inventory and booking capabilities.
- Partner agencies can integrate using documented API interfaces.
- The project uses five working days per week.
- The given estimates represent expected engineering effort.
- The first five partners are sufficiently representative for initial onboarding.

11 Requirements Traceability

Requirement Description		Test Case
FR-01	Partner registration	TC-001
FR-02	Authentication	TC-002
FR-03	Hotel search	TC-003
FR-04	Availability	TC-004
FR-05	Booking creation	TC-005
FR-06	Booking retrieval	TC-006
FR-07	Cancellation	TC-007
FR-08	API key management	TC-008
FR-09	Rate limiting	TC-009–TC-012
FR-10	Sandbox	TC-013

Requirement Description		Test Case
FR-11	Usage reporting	TC-014
FR-12	Documentation	TC-015

12 Acceptance Criteria

The initial release shall be considered ready when:

1. The first five partners can authenticate successfully.
2. Hotel search operates through the documented API.
3. Booking operations work for supported scenarios.
4. Sandbox access is available.
5. Documentation is complete.
6. Rate limiting and access control tests pass.
7. Production API error rate target is below 0.5%.
8. Partner onboarding target remains below 5 working days per partner.

13 Conclusion

This SRS defines the baseline requirements for PartnerAPI and provides the foundation for architecture, UML design, implementation, testing, project planning and change management.

The requirements establish a controlled initial release focused on common partner needs while keeping partner-specific custom requests outside the standard release scope unless approved through formal change control.